AUREX

PRODUCT ASSURANCE

Product quality testing summary

U-100 INSULIN SYRINGE

31G | 8 mm

Precision needle specification
31 gauge cannula
8 mm needle length



THIRD-PARTY TESTING

PASSED

AUREX reports that its U-100 insulin syringes, 31G x 8 mm, passed the third-party tests and evaluations listed in this summary.

Product specifications

Sterilization	**Ethylene Oxide (EO)**
Shelf life	**5 years**
Latex	**Latex free**
Substance exclusions	**PHT / DEHP / PVC / BPA free**

Document basis: Specifications and third-party pass results were reported by AUREX. Original laboratory reports, test methods, acceptance limits and lot identifiers were not supplied for preparation of this summary. This document does not replace the laboratory reports.

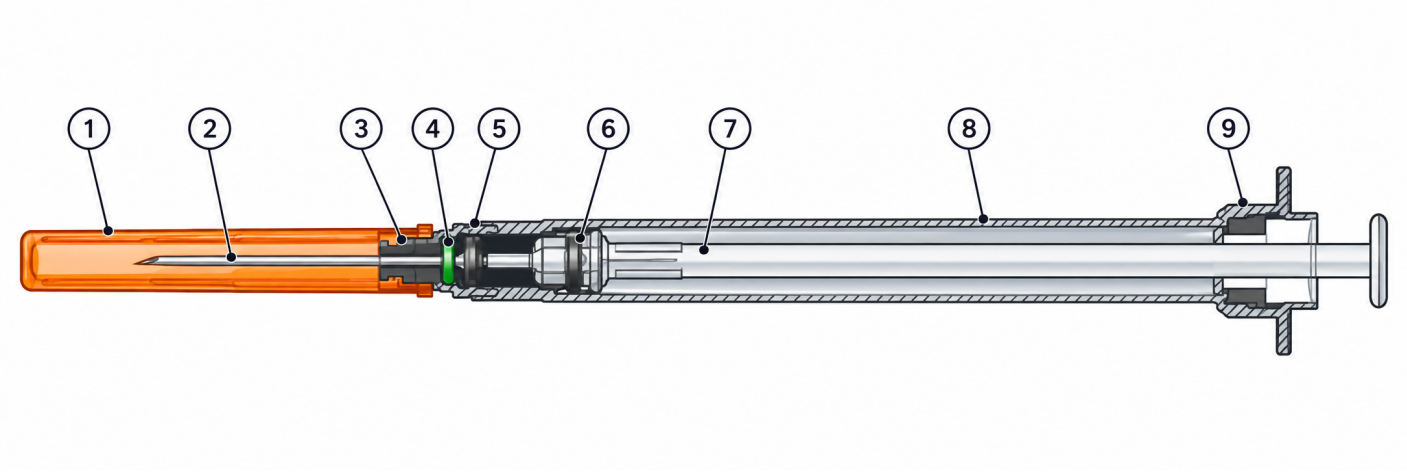
Testing & evaluation results

U-100 INSULIN SYRINGE / 31G x 8 mm

TEST / EVALUATION	ASSESSMENT	RESULT
Needle dimensions	Needle length and diameter	**PASS**
Needle-point inspection	Tip condition and visible defects	**PASS**
Penetration force	Resistance during standardized penetration	**PASS**
Needle-to-hub bond strength	Security of the needle attachment	**PASS**
Needle leakage	Leakage at the needle connection	**PASS**
Plunger-seal leakage	Leakage past the plunger gasket	**PASS**
Plunger fit and operation	Fit and functional movement in the barrel	**PASS**
Graduated-capacity accuracy	Volume relative to the graduation scale	**PASS**
Dead-space volume	Residual fluid after plunger depression	**PASS**
Lubrication assessment	Applicable lubrication requirements	**PASS**
Particulate contamination	Particles and foreign matter	**PASS**
Fragmentation	Fragments generated during closure piercing	**PASS**
Acidity / alkalinity of extracts	Chemical characteristics of device extracts	**PASS**
Extractable metals	Metals released under test conditions	**PASS**
Cytotoxicity	Cell response to device extracts	**PASS**
Sensitization	Potential for an allergic response	**PASS**
Irritation / intracutaneous reactivity	Potential for local tissue irritation	**PASS**
EO sterilization validation	Sterilization process performance	**PASS**
EO / ECH residuals	Residual sterilant and reaction product	**PASS**
Packaging integrity	Integrity of the sterile barrier	**PASS**
Packaging strength	Resistance to specified package stresses	**PASS**
Shelf-life evaluation	Aging-related performance and integrity	**PASS**

Designed as a complete system

12 COMPONENTS & ASSEMBLY MATERIALS



Illustrative cutaway redrawn from the supplied reference; not to scale. Numbers 1-9 identify the illustrated parts. Adhesive and lubricants are described below.

REF.	COMPONENT	FUNCTION
01	**Needle cap**	Protective cover over the needle before use.
02	**Cannula**	Hollow needle that provides the fluid pathway; 31G x 8 mm.
03	**Needle hub**	Supports the cannula at its connection to the syringe.
04	**O ring**	Ring-shaped sealing element within the illustrated assembly.
05	**Locating ring**	Positions and aligns mating parts within the assembly.
06	**Gasket**	Sealing element at the interface of the plunger and barrel.
07	**Plunger**	Moving element used to draw fluid into and expel it from the barrel.
08	**Barrel**	Fluid chamber carrying the U-100 insulin graduation scale.
09	**Locking ring**	Retaining element that helps secure the illustrated assembly.
10	**Adhesive**	Bonding material used to secure designated assembly interfaces.
11	**Needle lubricant**	Surface lubricant intended to reduce resistance during insertion.
12	**Barrel lubricant**	Lubricant intended to support smooth plunger movement.

Component functions are descriptive interpretations of the supplied schematic. Exact materials, geometry and assembly details are governed by the manufacturer's technical documentation.